Multi-Growth-Stage Pest Detection in Field Images using Deformable DETR with Self-Supervised Learning

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What?

We developestsp an SSL-DefDETR framework to detect and classify agricultural across multiple growth stages in field images.

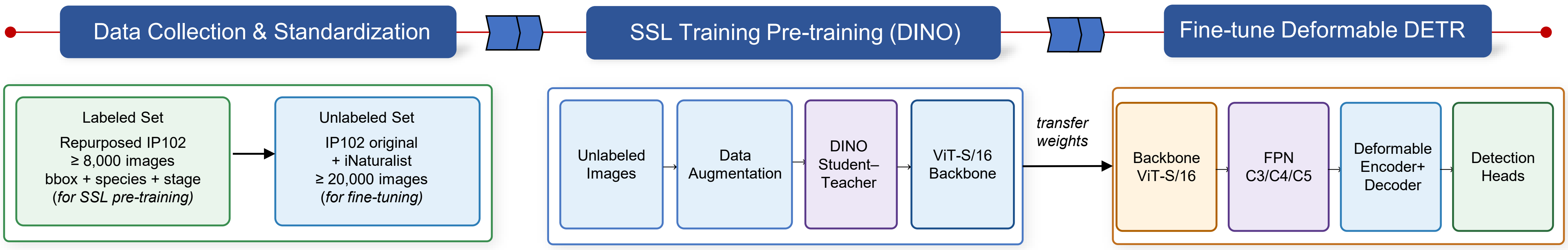
- Repurposed IP102 for species and growth-stage labels.
- Self-supervised pre-training with DINO on large-scale unlabeled images.
- Deformable DETR with SSL-pretrained backbone for detection.
- Joint prediction: species, growth stage, and bounding box.
- Improved performance under limited labeled data.

Why?

- Agricultural pests cause 20–40% crop loss worldwide (FAO).
- Early detection is critical for precision agriculture.
- Pest appearance changes dramatically across growth stages.
- Field images: complex backgrounds, variable illumination.
- Small-object detection remains a significant challenge.
- Labeled datasets are scarce and expensive to collect.

→ Limited labeled data and multi-stage pest variation motivate the use of SSL with Deformable DETR

OVERVIEW



DESCRIPTION

① Dataset Preparation

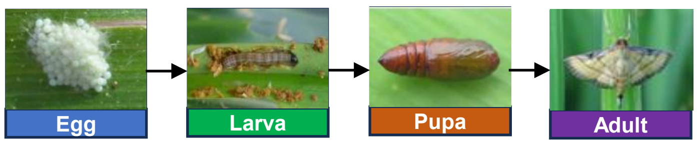


Figure 1. Pest Growth Stages Example: Fall Armyworm (FAW)

Labeled Set — Repurposed IP102

- Filter species with ≥ 2 labeled growth stages.
- SAM2 (zero-shot) generates object masks and bounding boxes
- Manual quality check on random 10% of images.
- Split: Train / Val / Test = 70% / 15% / 15%.

Labeled Set Construction

Item	Value
Total Images	≥ 8,000
Pest Species	≥ 10
Growth Stages	4 (Egg, Larva, Pupa, Adult)
Split	70% / 15% / 15%

Unlabeled Set for SSL

- IP102 original (unlabeled, ~60,000 images).
- iNaturalist field insect images.
- ≥ 20,000 images total for SSL pretraining.

② SSL Pre-Training (DINO)

- Unlabeled pest images from IP102 and iNaturalist.
- Multi-crop augmentation with crop, color, and noise transformations.
- Student–Teacher self-distillation (DINO).
- ViT-S/16 backbone for feature representation learning.
- Representation quality assessment via Linear Probe and kNN.
- Transfer SSL-learned representations to improve pest detection with limited labeled data.

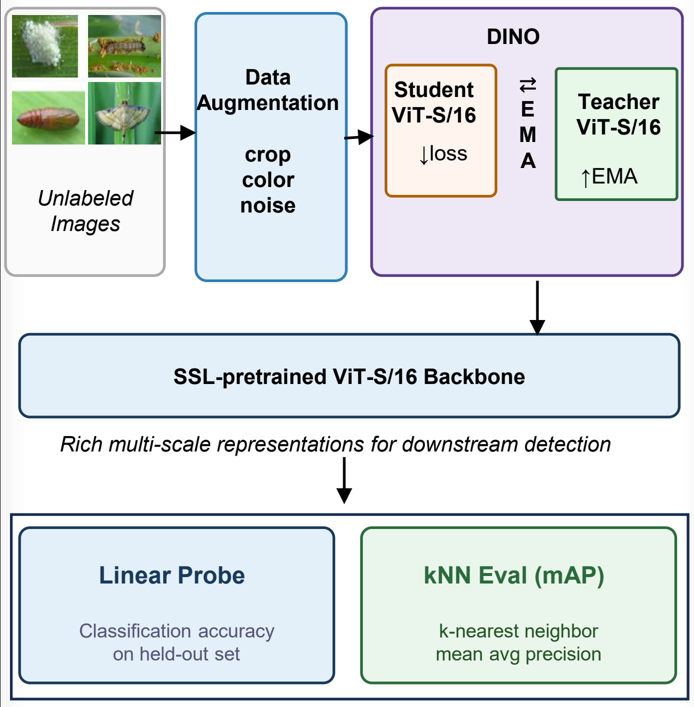


Figure 2. DINO self-supervised learning framework and representation evaluation.

③ SSL-DefDETR Fine-Tuning

- Initialize backbone with SSL-pretrained ViT-S/16 weights.
- Extract multi-scale features via FPN adapter (C3, C4, C5).
- Deformable encoder–decoder with MSDeformAttn.
- Jointly predict: species, growth stage, bounding box.
- Optimize using Hungarian matching, classification, L1, and GIoU losses.

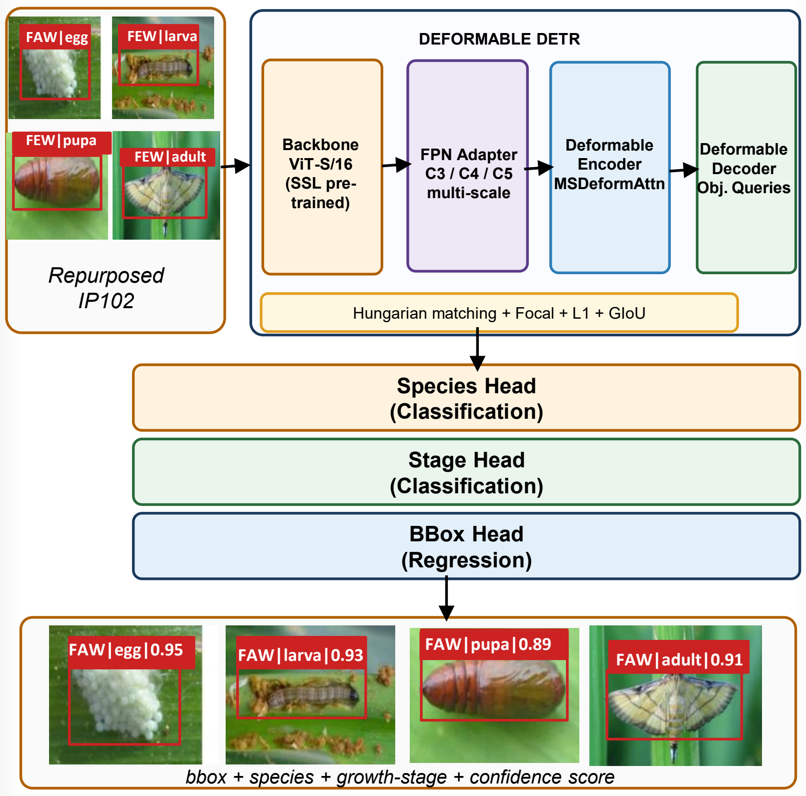


Figure 3. SSL-DefDETR architecture with multi-task prediction heads.

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